

Dear Sir/Ma'am

As a student of Mathematics and Computer Science at the University of Warwick - with a foundation in software and algorithm development, I am applying for the Software Engineering Intern position.

I enjoy problem-solving, and am passionate about building clean, efficient systems which scale easily and can have real-world impact. I am curious how I would apply such analytical skills to a real-world system such as trading and financial markets.

I look forward to speaking with you in further detail about the open position and opportunity with Optiver..

Sincerely,
Aarav

Advertised Requirements	My Qualifications
Passionate for clean code, simple, well-architected systems and continuous improvement	I practice writing clean code that scales, when I try to design my own websites, such as an AI chatbot that will automatically generate TMUA papers for you.
Proficient in industry-standard coding languages such as C++, Python, or Java	Through competitive programming, I have built a proficient use of languages such as C++ and Python, and through my degree I have developed my skills in Java as well. I have enriched these skills by doing hybrid projects outside of university, such as a program to generate and then compute the weights of arbitrary complete non ambiguous trees.
Strong understanding of underlying computer architecture and system operations	I have built this skill throughout my degree, and I have applied this outside of university by building a computer in Minecraft and/or building my own virtual machine in Rust.
An analytical mindset and strong problem-solving skills	I have demonstrated this through the various competitions I have done throughout my time. In addition my Mathematics degree has pushed me to focus on formal logic, which greatly helped with my analytical approach to problems.
Have an entrepreneurial attitude and an interest in the financial industry	I'm fascinated by the world of quantitative trading, participating in Kaggle Competitions and challenges like Quant Challenge 2025. I enjoy starting up projects, such as the TMUA AI paper generator, which I gave to my students at the time of tutoring them.
Able to work closely with other engineers, traders and business operations	I have worked in collaborative environments before, such as when building the robot to automatically play Connect 4 and adapt to your emotions. I also have an eagerness to learn from senior engineers and traders